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GEOTECHNICAL ENGINEERING

Module coursework, 2022-2023

EC7 Design and Plaxis modelling challenge

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1. Problem definition

The problem at hand is a temporary excavation trench, located in the middle of St George's Square Mews in London. The trench is made up of two opposite sheet pile walls and a steel props inbetween for support (see Figure 1). The analysis of this construction project requires consideration of various parameters, such as the depth of the excavation, the dimensions of the trench, external loading and the soil properties. A numerical analysis of the problem with reasonable assumptions will be presented, explained and compared to a model, made using the geotechnical engineering software Plaxis. An analysis of this trench will include the following outputs: the earth pressure or effective stress, the bending moments and the lever arms. One of the main challenges of this analysis is ensuring the stability and safety of the excavation trench after construction, as well as minimising any potential impact on the surrounding structures and environment. Another challenge is to optimise the design to minimise the cost and duration of the project while meeting the required safety and quality standards.

1.1 Parameters

1.1.1 Geometry

The dimensions of the geometry for the excavation trench are shown in Table 1. The values were set by the University of Greenwich, the client.

Table 1. Geometry of excavation trench

Variable name in Figure 1	Dimension Name	Value
a	Building storeys	3 storeys
b	Building length	8m
c	Machinery loading length	3m
d	Steel props length	2.5m
e	Water Table height	1.5m
f	Excavation height	3.5m

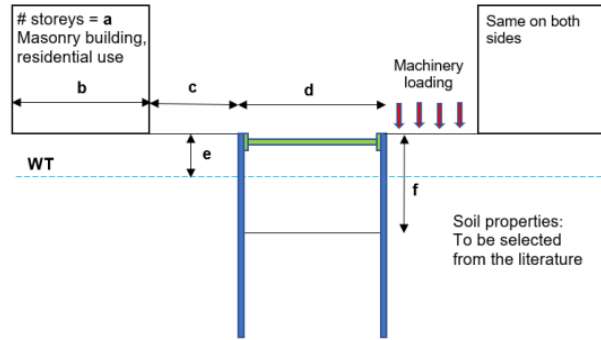


Figure 1. Geometry of excavation

1.1.2 Loading Parameters

Table 2. Loading

Load type	Load (kN)
Dead load from building	23.5
Live load from building	21
Machinery loading	21

Dead load can be calculated using the following formula

$$G_k = \gamma \cdot V \quad (1)$$

Where G_k is the dead load (kN), γ is the unit weight of the material of the structure (kN/m^3) and V is the volume of the structure (m^3).

The structure is defined as a masonry building so it can be assumed that brick, stone and concrete blocks were used in the construction (Srivastava, 2019). $19.2kN/m^3$ was used as the unit weight of the bricks (Krishna, 2019). With the assumption of a 8 by 8 base and a average UK floor height of $2.7432m$ (Lefton & Vila, 2019). Using equation 1 the dead load of the building can be found as $10,071kN$. However this value was an unrealistic value so a more realistic value was taken from Eurocode 7 Geotechnical Design Worked examples (Bond et al., 2013) the dead load for this building is 23.5 kN (see Table 2).

The live load of the building can be calculated by taking values from Table 6.2 presented in Eurocode 1 (BS EN 1991-1-1, 2002), Table 6.2 shows the range of live loading values given by the National annex, the building is for residential use so it falls into category A (see Appendix E-1). Table 3 shows the chosen live loads for the components in the building.

Table 3. Live loads for loaded areas in building

Loaded Areas	$Q_k (kN)$
Floors	3
Stairs	4
Roof	0

With an assumption of 0 kN live load on the roof and the assumption of one set of stairs per floor the live load was calculated to be 21 kN .

Using BS EN 1991-1-1 it was assumed that the machinery would have similar dimensions to the Forklift class FL1, in BS EN 1991-1-1 Table 1 the net weight is provided as 21 kN which was used as the machinery loading for this excavation.

1.1.3 Soil Parameters

Soil was assumed to be a typical thanet sand soil, sandy silt (uniform and fine grained). Although the excavation is located in London, the use of London clay was avoided due to the high cohesion which falls outside of the scope of this project. Soil properties were taken from research done by ABG Geosynthetics and GOV.UK (ABG, 2021) (See Table 4).

Table 4. Soil Properties

Soil property	Soil property values
Effective cohesion c'	3 kPa
Unit weight γ_{soil}	20.5 kN/m^3
Effective friction angle ϕ'	30°

2 Problem analysis

This section will be presenting the details of the analysis, assumptions for the actions and formulas used for the analysis of stability and safety of the excavation trench. This section is split into five sections: surcharge, geotechnical stability analysis, Design Approach 1 Geotechnical Design (DA1 GEO), Hydraulic ultimate limit state (HYD ULS) and structural design ultimate limit state (STR ULS)

2.1 Stability analysis: pile sheet

Due to the symmetrical nature of the problem one side of the problem can be analysed and then applied to the other side (see Figure 1).

2.1.1 Surcharge

To satisfy the scope of this project, the building loading and the machinery loading were treated as a singular structural load at the top of the trench, so the combined variable loads were found using BS EN 1990 to get the surcharge. Below are simplified versions of equation 6.10a and 6.10b equations for combined variable loads with varying favorability (BS EN 1990, 2002).

$$1.35G_k + 1.50\psi_0 Q_{k,1} + 1.50\varphi_0 Q_{k,2} \quad (2)$$

$$1.15G_k + 1.50\psi_0 Q_{k,1} + 1.50Q_{k,2} \quad (3)$$

$$1.15G_k + 1.50Q_{k,1} + 1.50\varphi_0 Q_{k,2} \quad (4)$$

Where ψ_0 is 0.7 taken from BS EN 1990 Table A1.1 Recommended values for ψ factors for buildings (BS EN 1990, 2002). The surcharge was found to be 80.575 kN

2.1.2 Analyse the stability of the embedded retaining walls

This section is broken down into steps used to analyse the problem (see Appendix D for full calculations).

Step 1: Water pore pressure

$$u = g \cdot z \quad (5)$$

Where u is pore water pressure (N/m^2), g is the acceleration of gravity (m/s^2) and z is the depth below the water table (m).

Step 2: Vertical stress

$$\sigma_v = \gamma_{soil} \cdot z + q \text{ (with surcharge, active side)} \quad (6)$$

$$\sigma_v = \gamma_{soil} \cdot z \text{ (passive side)} \quad (7)$$

Where σ_v is vertical stress (kN/m^2) and z is the depth (m).

Step 3: Effective vertical stress, σ'_v

$$\sigma'_v = \sigma_v - u$$

Step 4: Find K_a and K_p

$$K_a = \frac{1 - \sin\phi'}{1 + \sin\phi'} \quad (8)$$

$$K_p = \frac{1 + \sin\phi'}{1 - \sin\phi'} = \frac{1}{K_a} \quad (9)$$

Where K_a is the active earth pressure coefficient (dimensionless) and K_p is the passive earth pressure coefficient (dimensionless)

Step 5: Effective horizontal stress

$$\sigma'_h = K_a \sigma'_v - 2c'\sqrt{K_a} \text{ (active side)} \quad (10)$$

$$\sigma'_h = K_p \sigma'_v + 2c'\sqrt{K_p} \text{ (passive side)} \quad (11)$$

Where σ'_h is the effective horizontal stress (kN/m^2).

Step 6: Divide the effective horizontal stress into areas, find thrust force and lever arms

Lever arm is the point where the maximum thrust force is being applied. The point of rotation for the structure is assumed to be at the top of the excavation by the anchor or steel prop so the lever arm is the distance from the step prop to the middle of the area where the maximum force is applied.

$$P = \frac{1}{2} \cdot z \cdot \sigma'_h \quad (12)$$

Where P is the thrust force (kN).

Step 7: Maximum thrust force

$$P_{max} = \frac{k_p \gamma d^2}{2} \quad (13)$$

Where d is the lever arm for the passive side (m).

Step 8: Mobilising thrust force

$$P_{mob} = \frac{\sum P_a^1 \cdot d_a^1 \cdot P_a^2 \cdot d_a^2 \cdot \dots}{P_{max} \cdot I_p} \quad (14)$$

Where P_a^1 is the thrust force for each area and d_a^1 is the lever arm of the corresponding area.

Where I_p is equal to the level arm on the passive side (m).

Step 9: Factor of safety

$$FoS = \frac{P_{max}}{P_{mob}} \quad (15)$$

2.2 DA1 GEO ULS

Eurocode 7 introduces the idea of limit state analysis and design for geotechnical problems. Limit states are states beyond which the structure no longer satisfies the relevant design criteria, the ULS applies to all possible failure modes of a structure. In BS EN 1997-1 there are 5 limit states, the GEO state is to be considered which is failure in ground or excess deformation of the ground this includes: bearing failure, forward sliding and ground failure (BS EN 1997-1, 2004).

Design approach one has the following combinations:

$$\text{Combination 1: } A1 + M1 + R1 \quad (16)$$

$$\text{Combination 2: } A2 + M2 + R1 \quad (17)$$

Where A is the design values of actions, M is design values of the geotechnical parameters and R is the design value of resistance.

For the specified problem only A and M will be needed here are how each of the values were changed based on the combination case.

For A :

$$F_d = F_{rep} \cdot \gamma_F \quad (18)$$

Where F_d is the design value of action F , F_{rep} is the representative value and γ_F is the partial factor of safety found in Table A.3 in Eurocode 7 (see Appendix G).

For M :

$$X_d = \frac{X_k}{\lambda_M} \quad (19)$$

Where X_d is the design value of geotechnical parameter X , X_k is the characteristic value and λ_M is the partial factor of safety found in Table A.4 in Eurocode 7 (see Appendix G).

2.3 HYD ULS

HYD ULS is a limit state defined in Eurocode 7 for the design of geotechnical structures subjected to hydraulic actions such as water pressure or flow. This limit state considers the possibility of failure due to hydraulic loads. Hydraulic upheave failure occurs when soil or other materials beneath a structure experience an increase in pore water pressure due to the presence of water or other fluids. This can lead to the upward movement of the soil or materials, causing damage or failure of the foundation.

Equations BS EN 1997-1 equations 2.9a and 2.9b respectively, are used to verify there will be no uplift heave in the wall .

$$u_{dst;d} \leq \sigma_{stb;d} \quad (20)$$

Where $u_{dst;d}$ is the water pore pressure and $\sigma_{stb;d}$ is the total vertical stress at the bottom of the sheet pile.

$$S_{dst;d} \leq G'_{stb;d} \quad (21)$$

Where $S_{dst;d}$ is the design value of seepage force and $G'_{stb;d}$ is the stabilising vertical stress at the bottom of the sheet pile. See Appendix D-2 for full calculations.

2.4 STR ULS

STR ULS is a limit state defined in Eurocode 3 for the design of steel structures. The STR ULS requires that the steel structure be designed to withstand the maximum load that it may be subjected to during its lifetime without experiencing structural failure or unacceptable levels of deformation. This limit state considers the strength and stability of the structure, as well as the ductility and resistance to fatigue under cyclic loading. The design must satisfy the requirements of the STR ULS to ensure the safety and durability of the steel structure. The b , breadth (m) and t_f , the thickness of flange in the chosen pile wall must satisfy equation 22 below (BS EN 1993-5, 2007)

$$\frac{b/t_f}{\varepsilon} \leq 37 \quad (22)$$

$$\varepsilon = \sqrt{\frac{235}{f_y}} \quad (23)$$

Where f_y is the yield strength of steel 270 N/mm^2 .

A U shaped pile walling was chosen, ArcelorMittal GU 33N sheet piling was chosen as it satisfies the STR ULS check performed and it is the most cost effective for 9m length of the wall. See Appendix D-3 for full calculations.

3 Analysis results & Discussion

During the analysis of the sheet pile, the pile was divided into 6 areas on the active side (right) and 1 area on the passive side (left) (see Figure 2).

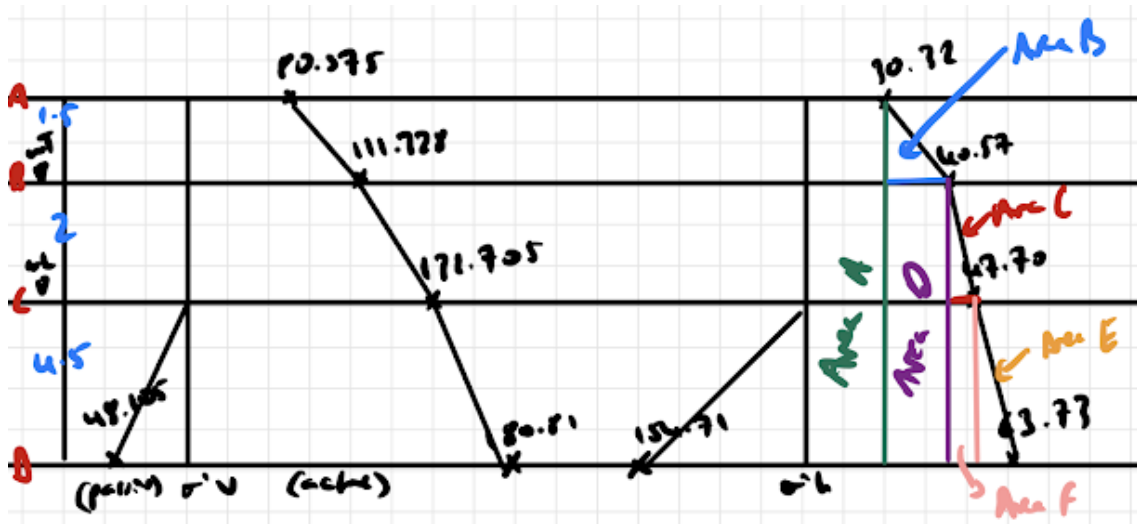


Figure 2. Abstract from Appendix D-1

Important results of the analysis include the factor of safety, mobilising and maximum thrust force maximum and moment per metre of pile sheet. In geotechnical engineering, assuming the thrust force acts perpendicular to the wall the moment can be calculated by multiplying the thrust force by the lever arm.

Table 5. Analysis results hand calculations (normal analysis)

Area	Thrust force (P)	Lever arm	Moment (M)	Moment per metre
	kN	m	kNm	kNm/m
A (active)	105.2740427	4.5	473.7331923	52.63702137
B (active)	7.6875	1	7.6875	0.854166667
C (active)	7.126666667	2.83333333	20.19222222	2.243580247

D (active)	38.4375	5.25	201.796875	22.421875
E (active)	53.89541667	7.1666667	386.2504861	42.91672068
F(active)	19.59833333	6.25	122.4895833	13.6099537
Passive	494.5850294	7.1666667	3544.526044	393.8362271
Factor of Safety		25.38167222		

Table 5. Analysis results hand calculations (GEO DA1 Combination 1)

Area	Thrust force (P)	Lever arm	Moment (M)	Moment per metre
	kN	m	kNm	kNm/m
A (active)	159.6247	4.5	718.311	79.81233387
B (active)	7.6875	1	7.6875	0.854166667
C (active)	7.126667	2.8333333	20.19222	2.243580247
D (active)	38.4375	5.25	201.7969	22.421875
E (active)	53.89542	7.1666667	386.2505	42.91672068
F(active)	19.59833	6.25	122.4896	13.6099537
Passive	494.585	7.1666667	3544.526	393.8362271
Factor of Safety		24.14042		

Table 6. Analysis results hand calculations (GEO DA2 Combination 2)

Area	Thrust force (P)	Lever arm	Moment (M)	Moment per metre
	kN	m	kNm	kNm/m
A (active)	169.3748	4.5	762.1868	84.68742
B (active)	9.726153	1	9.726153	1.080684
C (active)	9.016592	2.8333333	25.54701	2.838557
D (active)	48.63077	5.25	255.3115	28.36795
E (active)	68.18798	7.1666667	488.6805	54.29783
F(active)	24.79563	6.25	154.9727	17.21919

Passive	391.96	7.1666667	2809.047	312.1163
Factor of Safety		16.74062		

Table 7. HYD check results

HYD check 1	
$S_{dst;d}$	3.996211
$G'_{stb;d}$	8.829
$S_{dst;d} \leq G'_{stb;d}$	TRUE - accept
HYD check 2	
$u_{dst;d}$	94.81841
$\sigma_{stb;d}$	97.119
$u_{dst;d} \leq \sigma_{stb;d}$	TRUE - accept

STR check was fulfilled see Appendix D-3.

4 Plaxis modelling

4.1 Modelling parameters and assumptions

4.1.1 Boundaries

For modelling in Plaxis 2D a point of origin 0,0 needs to be established in order to add the individual components to the problem. First the entirety of the problem or the boundaries needed to be established. In the y-direction it was assumed to be unimportant as the focus was on the horizontal stresses and moments on the pile sheet, so the y-maximum was double the pile sheet. Additionally good geotechnical practice dictates that the x-maximum in regards to retaining walls problems, be five times the length of the components along the ground level. Figure 3 shows the point of origin in green at the bottom left.

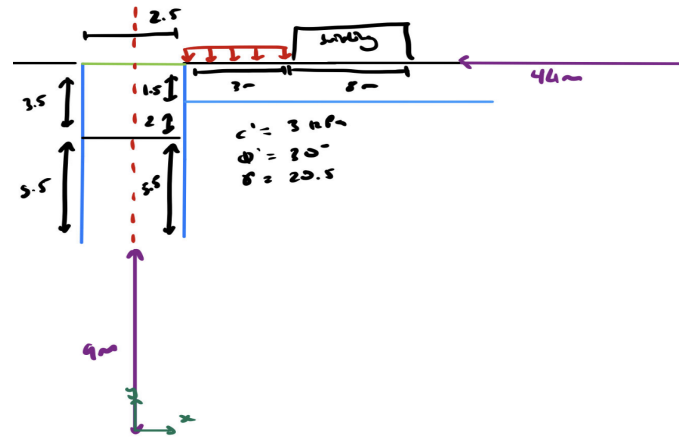


Figure 3. Sketched Plaxis model with of the boundaries and point of origin

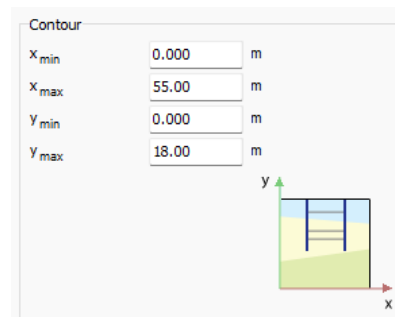


Figure 4. Shows the boundary for the model inputted into Plaxis 2D

4.1.2 Steel props

As mentioned in section 2. Due to the symmetrical nature of the question only one side was modelled, in order to simulate the steel props between two sheet pile walls, a fixed end anchor was added, the restraint had a length of half the length of the steel props so 1.25m (see Figure 3).

4.1.3 Loading used in modelling

The load of the machinery used in Plaxis was 21kN. The total load of the building was taken from equation 24.

$$\gamma_G G_k + \gamma_Q Q_k \quad (24)$$

Where γ_G is the ULS partial safety factor of permanent actions 1.35 and γ_Q ULS partial safety factor of variable actions 1.5. Where the building live and dead loads are taken from Table 2. The total load on the building used in Plaxis was 63.225kN

4.1.4 Soil Properties

Table 8. Soil properties used in Plaxis

Soil property	Value
Drainage Type	Undrained A
Unit weight unsaturated γ_{unsat}	18.5 kN/m^3
Unit weight saturated γ_{sat}	20.5 kN/m^3
Stiffness E	$30,000 \text{ kN/m}^2$
Passion rations ν	0.2 <i>nu</i>

Values in this table 8 were used with the values displayed in section 1.1.3 during Plaxis modelling of soil properties.

4.1.5 Piling wall assumptions

In conjunction with the checks done in section 2.4 STR ULS. Arcelormittal GU 33N sheet pile was used. Table 9 is a design table used for the piling wall in Plaxis, steels elastic modulus is 210 GPa .

Table 9. Plaxis properties of sheet pile wall

Type	Weight [KN/m/m]	Axial stiffness [kN/m]	Flexural rigidity [kNm^2/m]
GU 33N	0	$5026 \cdot 10^3$	150820

The weight of the piling wall was not important for this analysis so it was assumed to be negligible. The equations 25 and 26 were used to find the values in the design Table 9, using the ArcelorMittal sectional property table for sheet piles (see Appendix G).

$$\text{Axial stiffness} = E \cdot A \quad (25)$$

$$\text{Flexural rigidity} = E \cdot I \quad (26)$$

The flexural rigidity of the anchor was assumed to be equal to that of the sheet pile.

4.2 Procedure

The following section briefly describes the procedures used to create the model in Plaxis. First, the point of origin and the working area of the model was established. Then in the soil tab of Plaxis, the borehole tool was used to define the soil layer. The soil layer was modified to create the water table at 16.5m. Next in the structures tab, the plate element tool was used to draw the sheet pile 9m long, and a fixed end anchor was added to simulate the steel

prop. The loading from machinery and building was added with respect to the distances shown in table 2. An excavation cluster was created, and the excavation was split into two parts. The interface tool was used to create an interface around the sheet pile, and the soil materials were edited using information from tables 4. A plate element was created for the wall and anchor, and the properties shown in section 1. On the hash tab section, the sheet pile was refined so that Plaxis can get more accurate results. On the flow conditions tab, a secondary water table was added to simulate the water level after excavation. Finally, the analysis was run on the stage construction tab, activating all important information such as loads, anchors, and excavation area. The analysis was split into different sections, including the initial phase, excavation at different depths, and dewatering.

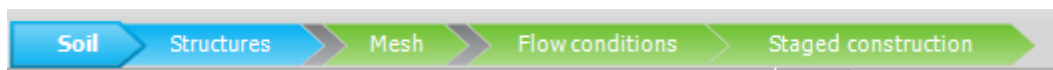


Figure 5. Plaxis design tabs

5 Numerical results & Discussion

5.1 Numerical result table and diagrams

Table 10: Comparison of results Plaxis vs Numerical

Name	Plaxis results	Numerical results
σ'_h active side max	-79.40	63.73
σ'_h active side min	-2.98	30.32
σ'_h passive side max	-121.27	154.71
σ'_h passive side min	6.97E-07	0
Bending Moment max	24.18	3544.526044
Bending Moment min [kNm /m]	136	7.6875
Shear Force max [Kn/m]	53.23	N/A
Shear Force min [Kn/m]	-88.90	N/A

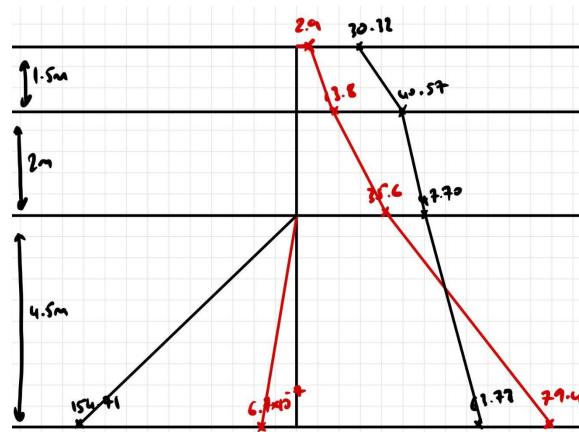


Figure 6. Sketched comparison of earth pressure. Black is handwritten and red is Plaxis

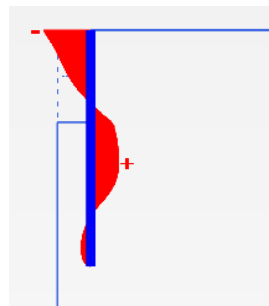


Figure 7. Plaxis shear force diagram of the trench

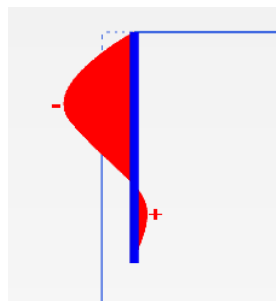


Figure 8. Plaxis bending moment diagram of the trench

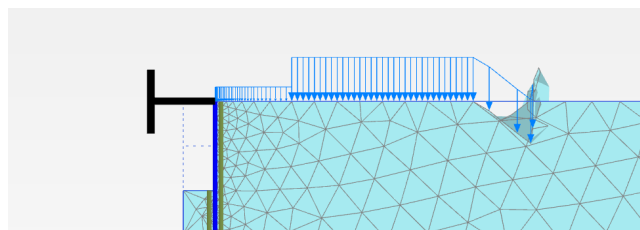


Figure 9. Plaxis deformed mesh, maximum deformation 4.579E9 (scaled up 0.5E-9)

5.2 Discussion

The differences between the numerical calculations and the outputs from the Plaxis model can be attributed to multiple factors. Firstly, in the numerical calculations, the forces were assumed to act towards the wall without a global system, while Plaxis applied a global system. Moreover, in the Plaxis model, the machinery and building loads were considered separately, while they were put together in the numerical calculation represented as a singular surcharge (see section 2.1.1). Additionally, the larger differences in the moment on the passive side is most likely due to human error. Finally, there is an extreme settlement of about 2.2895m (see Figure 9) which is an interesting finding, most of the deformation occurs on the rear of the residual building which is useful information for design refinements.

6 Serviceability checks and design refinements

Based on the displacement values from Plaxis the horizontal boundary was increased, which caused less deformation, the loading was also slightly adjusted which caused for the deformation to form at the anchor which was expected and refinements were added to the top of the wall (see Appendix C-5).

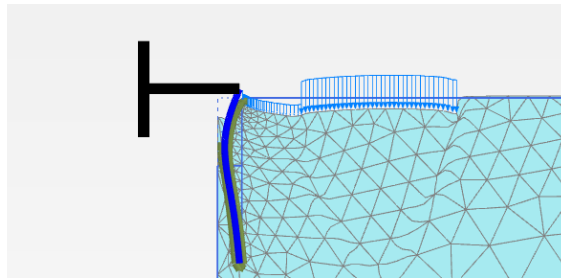


Figure 10. Plaxis adjusted model deformed mesh, maximum deformation 5.005E-3 (scaled up 500 times)

7 Conclusions

In conclusion, the findings suggest that there are errors in the assumptions and methods, which contributed to the differences between the numerical calculations and the Plaxis model outputs. The numerical calculations had numerous assumptions, particularly regarding the loading combinations used to treat the machinery and building at the surface as one surcharge, which may have limited the results. Additionally, the method used to calculate the total loading of the building used in Plaxis may also be flawed. On the other hand, Plaxis had a multitude of additional considerations and assumptions to increase accuracy, such as soil properties and specific sheet pile design specifications, which can significantly affect the results. Conclusively human errors in calculations are the primary reasons for differences in results.

To address these issues, several changes are suggested. Firstly, there should be a reevaluation of the analysis used to find the Factor of Safety using this an optimal sheet pile length should then be selected. Secondly, the modelling boundary conditions should be increased to better mimic the location of the trench, decreasing settlement differences. Finally, it is recommended that shear forces be verified using numerical evaluations. By implementing these changes, more accurate and reliable results can be obtained, leading to better decision-making in the project design. It is essential to ensure that assumptions and limiting factors are minimised to improve the accuracy of the calculations and models, thereby improving the overall success of the project. In the engineering industry numerical analysis and hand calculations are peer reviewed to minimise mistakes before modelling takes place. Furthermore more research is required before an accurate final design proposal can be offered.

Reflecting on the project, it was highly stimulating and challenging, encompassing multiple components and considerations such as geotechnical aspects, hydraulic aspects, stability of structures, design, and modelling. This project effectively highlighted the diversity of civil engineering and the complex skill set required for success in this field. It also reinforced the significance of a keen eye for detail and the importance of minimising errors in numerical evaluations.

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Appendices

Appendix A: Excel spreadsheets

Appendix A-1: GEO normal stability

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	a	3 storeys			c'	3 kPa			Gk	23.5 kN			EC1 roof	0 kN	
2	b	8 m			gamma_soil	20.5 kN/m3			Qk1 (machine)	21 kN			EC1 floor	3 kN	
3	c	3 m			phi'	30 degrees			Qk2 (house)	21 kN			EC1 stairs	4 kN	
4	d	2.5 m			q	80.575 kN			distance A	0 m			psi0	0.7	
5	e	1.5 m			Ka	0.33333333			distance AB	1.5 m			ULS		
6	f	3.5 m			Kp	3			distance BC	2 m			ULS1	75.825 kN	
7	Pile	9 m		get FoS to phi (for excel)		0.523598776			distance CD	5.5 m			ULS2	80.575 kN	
8	g	9.81 m/s2											ULS3	80.575 kN	
9	Points/lines	phi' (degree)	c' (Kpa)	u	sigma v	eff sigma v	eff sigma h (active)	eff sigma h (passive)							
10	A	30	3	0	80.575	80.575	23.39423172								
11	B	30	3	0	111.325	111.325	33.64423172								
12	C	30	3	19.62	152.325	132.705	40.77089838								
13	D	30	3	73.575	265.075	191.5	60.36923172								
14	Left side	30	3	53.955	112.75	58.795		179.8491016							
15															
16	Areas	Pa	Pp	Lever arm		M		M/m							
17	Area A	105.2740427		4.5 m		473.7331923 kN/m2		52.63702137							
18	Area B	7.6875		1 m		7.6875 kN/m2		0.854166667							
19	Area C	7.126666667		2.8333333 m		20.19222222 kN/m2		2.243580247							
20	Area D	38.4375		5.25 m		201.796875 kN/m2		22.421875							
21	Area E	53.89541667		7.1666667 m		386.2504861 kN/m2		42.91672068							
22	Area F	19.59833333		6.25 m		122.4895833 kN/m2		13.6099537							
23	Area left side		494.5850294	7.1666667 m		3544.526044 kN/m2		393.8362271							
24	Ip	7.166666667													
25															
26	Ppmax	930.1875													
27	Pmob	36.64799908													
28	FoS	25.38167222													

Appendix A-2: GEO DA1 combination 1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	a	3 storeys			c'	3 kPa			Gk	31.725 kN			EC1 roof	0 kN	
2	b	8 m			gamma_s	20.5 kN/m ³			Qk1 (machine)	31.5 kN			EC1 floor	3 kN	
3	c	3 m			phi'	30 degrees			Qk2 (house)	31.5 kN			EC1 stairs	4 kN	
4	d	2.5 m			q	116.8088 kN			distance A	0 m			psi0	0.7	
5	e	1.5 m			Ka	0.333333			distance AB	1.5 m			ULS		
6	f	3.5 m			Kp	3			distance BC	2 m			ULS1	108.9788 kN	
7	Pile	9 m			phi (for ex	0.523599			distance CD	5.5 m			ULS2	116.8088 kN	
8	g	9.81 m/s ²											ULS3	116.8088 kN	
9	Points/lin	phi' (degr	c' (Kpa)	u	sigma v	eff sigma	eff sigma	eff sigma h (passive)	GEO 1	A1+M1+R1					
10	A	30	3	0	116.8088	116.8088	35.47215		G yG;dst	1.35					
11	B	30	3	0	147.5588	147.5588	45.72215		Q yG;dst	1.5					
12	C	30	3	19.62	188.5588	168.9388	52.84882		phi' yphi'	1					
13	D	30	3	73.575	301.3088	227.7338	72.44715		c' yc'	1					
14	Left side	30	3	53.955	112.75	58.795		179.8491016	y yy	1					
15															
16	Areas	Pa	Pp	Lever arm	M		M/m								
17	Area A	159.6247		4.5	718.311 kN/m ²		79.81233387								
18	Area B	7.6875		1	7.6875 kN/m ²		0.854166667								
19	Area C	7.126667		2.833333	20.19222 kN/m ²		2.243580247								
20	Area D	38.4375		5.25	201.7969 kN/m ²		22.421875								
21	Area E	53.89542		7.166667	386.2505 kN/m ²		42.91672068								
22	Area F	19.59833		6.25	122.4896 kN/m ²		13.6099537								
23	Area left side		494.585	7.166667	3544.526 kN/m ²		393.8362271								
24	Ip	7.166667													
25															
26	Ppmax	930.1875													
27	Pmob	38.53236													
28	FoS	24.14042													

Appendix A-3: GEO DA1 combination 2

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	a	3 storeys			c'	2.4 kPa			Gk	23.5 kN			EC1 roof	0 kN	
2	b	8 m			gamma_s	20.5 kN/m3			Qk1 (mach	27.3 kN			EC1 floor	3 kN	
3	c	3 m			phi'	24 degrees			Qk2 (hous	27.3 kN			EC1 stairs	4 kN	
4	d	2.5 m			q	96.64 kN			distance A	0 m			psi0	0.7	
5	e	1.5 m			Ka	0.42173			distance A	1.5 m			ULS		
6	f	3.5 m			Kp	2.371184			distance B	2 m			ULS1	89.055 kN	
7	Pile	9 m			phi (for e)	0.418879			distance C	5.5 m			ULS2	96.64 kN	
8	g	9.81 m/s2											ULS3	96.64 kN	
9	Points/lin	phi' (degr	c' (Kpa)	u	sigma v	eff sigma	eff sigma	eff sigma h (passive)	GEO 2	A2+M2+R1					
10	A	24	2.4	0	96.64	96.64	37.63885		G yG;dst	1					
11	B	24	2.4	0	127.39	127.39	50.60706		Q yG;dst	1.3					
12	C	24	2.4	19.62	168.39	148.77	59.62365		phi' yphi'	1.25					
13	D	24	2.4	73.575	281.14	207.565	84.41928		c' yc'	1.25					
14	Left side	24	2.4	53.955	112.75	58.795		142.5309	y yy	1					
15															
16	Areas	Pa	Pp	Lever arm		M		M/m							
17	Area A	169.3748		4.5		762.1868 kN/m2		84.68742							
18	Area B	9.726153		1		9.726153 kN/m2		1.080684							
19	Area C	9.016592		2.833333		25.54701 kN/m2		2.838557							
20	Area D	48.63077		5.25		255.3115 kN/m2		28.36795							
21	Area E	68.18798		7.166667		488.6805 kN/m2		54.29783							
22	Area F	24.79563		6.25		154.9727 kN/m2		17.21919							
23	Area left side		391.96	7.166667		2809.047 kN/m2		312.1163							
24	Ip	7.166667													
25															
26	Ppmax	735.2153													
27	Pmob	43.91804													
28	FoS	16.74062													

Appendix A-4: HYD ULS

	A	B	C	D	E	F	G
1	yG;dst	1.35		V (constar	1 m3		
2	yG;stb	0.9		y'	9.81		
3	yw	9.81 kN/m3		f	3.5 m		
4	h	3.5 m		Pile wall	9 m		
5	h'	2 m					
6	t	5.5 m		change size of pile sheet			
7	HYD CHECK 1					HYD CHECK 2	
8	hr	1.659619				uk	70.23586
9	i	0.301749				stb;k	107.91
10	Sk	2.960156				ukyG,dst	94.81841
11	G'k	9.81				stb;kyG,stb	97.119
12	skYG,dst	3.996211				ukyG,dst <= stb;kyG,stb	TRUE
13	G'kyGstb	8.829					
14	skYG,dst <= G'kyGstb		TRUE				

Appendix A-5: STR ULS

	A	B
1	Sheet pile name	GU 31N
2	b	600
3	tf	18.5
4	fy	270
5	e	0.932936
6	Class 2	34.76382
7	U-profile limit	37
8	Check	TRUE

Appendix A-6: Plaxis results part 1 portioned screenshot

	A	B	C	D	E	F	G	H	I
1	Y [m]	σ'_N [kN/m ²]	<--- active		Y [m]	σ'_N [kN/m ²]	<--- passive		
2	1.80E+01	8.62E-07			1.45E+01	-4.68E+01			
3	1.80E+01	-2.98E+00			1.45E+01	-4.80E+01			
4	1.79E+01	-6.96E+00			1.44E+01	-2.86E+01			
5	1.79E+01	-4.29E+00			1.44E+01	-5.96E+01			
6	1.79E+01	-9.17E+00			1.44E+01	-4.94E+01	max		-121.272617
7	1.79E+01	-9.17E+00			1.44E+01	-4.94E+01	min		6.96533E-07
8	1.79E+01	-4.77E+00			1.44E+01	-5.62E+01			
9	1.78E+01	-5.54E+00			1.43E+01	-5.32E-01			
10	1.78E+01	-5.79E+00			1.43E+01	-1.21E+02			
11	1.78E+01	-5.01E+00			1.43E+01	-2.05E+00			
12	1.78E+01	-5.01E+00			1.43E+01	-2.05E+00			
13	1.78E+01	-4.47E+00			1.43E+01	-4.60E+01			
14	1.77E+01	-4.61E+00			1.42E+01	-5.71E+01			
15	1.77E+01	-4.25E+00			1.42E+01	-4.71E+01			
16	1.77E+01	-4.40E+00			1.42E+01	-5.20E+01			

Appendix B: Plaxis results

**due to the refinements on the wall there are 4100+ nodes each sub-sections contains the results of one relevant structural element used in the completion of this project*

See next page

Appendix B-1: Passive effective horizontal stress

Structural element ▲	Node ▲	Wall node ▲	Local number ▲	X ▲ [m]	Y ▲ [m]	σ'_N ▲ [kN/m ²]	σ_N ▲ [kN/m ²]	τ_1 ▲ [kN/m ²]	τ_2 ▲ [kN/m ²]	τ_{rel} ▲ [10 ⁻³]	τ_{max} ▲ [kN/m ²]	P_{water} ▲ [kN/m ²]	P_{active} ▲ [kN/m ²]	Head ▲ [m]	P_{excess} ▲ [kN/m ²]	P_{steady} ▲ [kN/m ²]	Suction ▲ [kN/m ²]	Suction _{eff} ▲ [kN/m ²]	q_n ▲ [m/day]	Q_1 ▲ [m ³ /day/m]
NegativeInterface_1_3 Element 6-160 (Interface) (Soil)	5027	5031	1	1.250	9.647	-65.612	-151.029	3.482	0.000	116.402	29.911	-85.418	-85.418	18.189	-36.888	-48.529	0.000	0.000	0.000	0.000
	5014	5020	2	1.250	9.674	-65.284	-151.206	4.032	0.000	135.427	29.773	-85.922	-85.922	18.266	-37.662	-48.260	0.000	0.000	0.000	0.000
	5015	5019	3	1.250	9.701	-65.363	-151.378	4.578	0.000	153.604	29.806	-86.015	-86.015	18.302	-38.025	-47.990	0.000	0.000	0.000	0.000
	5016	5018	4	1.250	9.728	-65.429	-151.536	5.122	0.000	171.667	29.834	-86.108	-86.108	18.339	-38.387	-47.721	0.000	0.000	0.000	0.000
	5013	5017	5	1.250	9.755	-65.079	-151.692	5.663	0.000	190.745	29.686	-86.613	-86.613	18.416	-39.162	-47.451	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-161 (Interface) (Soil)	5063	5067	1	1.250	9.539	-66.384	-150.196	1.220	0.000	40.356	30.238	-83.811	-83.811	17.920	-34.203	-49.608	0.000	0.000	0.000	0.000
	5028	5034	2	1.250	9.566	-65.835	-150.420	1.797	0.000	59.893	30.006	-84.585	-84.585	18.025	-35.247	-49.338	0.000	0.000	0.000	0.000
	5029	5033	3	1.250	9.593	-65.877	-150.639	2.365	0.000	78.786	30.023	-84.763	-84.763	18.069	-35.694	-49.069	0.000	0.000	0.000	0.000
	5030	5032	4	1.250	9.620	-65.897	-150.837	2.927	0.000	97.447	30.032	-84.940	-84.940	18.114	-36.141	-48.799	0.000	0.000	0.000	0.000
	5027	5031	5	1.250	9.647	-65.612	-151.029	3.482	0.000	116.402	29.911	-85.418	-85.418	18.189	-36.888	-48.529	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-162 (Interface) (Soil)	5083	5087	1	1.250	9.431	-67.883	-149.087	-1.207	0.000	39.103	30.871	-81.204	-81.204	17.552	-30.518	-50.686	0.000	0.000	0.000	0.000
	5064	5070	2	1.250	9.458	-67.052	-149.390	-0.578	0.000	18.937	30.520	-82.338	-82.338	17.692	-31.922	-50.417	0.000	0.000	0.000	0.000
	5065	5069	3	1.250	9.485	-67.020	-149.687	0.035	0.000	1.140	30.506	-82.666	-82.666	17.752	-32.519	-50.147	0.000	0.000	0.000	0.000
	5066	5068	4	1.250	9.512	-66.949	-149.943	0.633	0.000	20.783	30.476	-82.994	-82.994	17.812	-33.117	-49.877	0.000	0.000	0.000	0.000
	5063	5067	5	1.250	9.539	-66.384	-150.196	1.220	0.000	40.356	30.238	-83.811	-83.811	17.920	-34.203	-49.608	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-163 (Interface) (Soil)	5549	5553	1	1.250	9.324	-70.887	-147.546	-3.958	0.000	123.150	32.140	-76.658	-76.658	16.989	-24.894	-51.765	0.000	0.000	0.000	0.000
	5084	5090	2	1.250	9.350	-69.205	-147.938	-3.226	0.000	102.653	31.429	-78.733	-78.733	17.224	-27.238	-51.495	0.000	0.000	0.000	0.000
	5085	5089	3	1.250	9.377	-69.042	-148.386	-2.527	0.000	80.576	31.361	-79.343	-79.343	17.312	-28.118	-51.225	0.000	0.000	0.000	0.000
	5086	5088	4	1.250	9.404	-68.785	-148.738	-1.856	0.000	59.380	31.252	-79.954	-79.954	17.400	-28.998	-50.956	0.000	0.000	0.000	0.000
	5083	5087	5	1.250	9.431	-67.883	-149.087	-1.207	0.000	39.103	30.871	-81.204	-81.204	17.552	-30.518	-50.686	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-164 (Interface) (Soil)	5573	5577	1	1.250	9.216	-75.950	-144.228	-7.363	0.000	214.786	34.279	-68.278	-68.278	16.043	-15.434	-52.843	0.000	0.000	0.000	0.000
	5550	5556	2	1.250	9.243	-73.686	-145.416	-6.421	0.000	192.683	33.322	-71.730	-71.730	16.416	-19.156	-52.574	0.000	0.000	0.000	0.000
	5551	5555	3	1.250	9.270	-73.322	-146.341	-5.544	0.000	167.156	33.168	-73.020	-73.020	16.572	-20.716	-52.304	0.000	0.000	0.000	0.000
	5552	5554	4	1.250	9.297	-72.585	-146.895	-4.727	0.000	143.871	32.857	-74.310	-74.310	16.728	-22.275	-52.034	0.000	0.000	0.000	0.000
	5549	5553	5	1.250	9.324	-70.887	-147.546	-3.958	0.000	123.150	32.140	-76.658	-76.658	16.989	-24.894	-51.765	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-165 (Interface) (Soil)	5884	5885	1	1.250	9.108	-84.639	-136.404	-12.044	0.000	317.375	37.949	-51.766	-51.766	14.284	2.156	-53.922	0.000	0.000	0.000	0.000
	5574	5580	2	1.250	9.135	-78.872	-138.071	-10.657	0.000	300.089	35.513	-59.199	-59.199	15.055	-5.547	-53.652	0.000	0.000	0.000	0.000
	5575	5579	3	1.250	9.162	-78.760	-140.533	-9.454	0.000	266.576	35.465	-61.773	-61.773	15.339	-8.391	-53.382	0.000	0.000	0.000	0.000
	5576	5578	4	1.250	9.189	-78.123	-142.471	-8.371	0.000	237.847	35.196	-64.348	-64.348	15.624	-11.235	-53.113	0.000	0.000	0.000	0.000
	5573	5577	5	1.250	9.216	-75.950	-144.228	-7.363	0.000	214.786	34.279	-68.278	-68.278	16.043	-15.434	-52.843	0.000	0.000	0.000	0.000
NegativeInterface_1_3 Element 6-166 (Interface) (Soil)	6052	5889	1	1.250	9.000	0.000	9.591	-0.668	0.000	304.251	2.195	9.591	9.591	8.041	64.591	-55.000	9.591	9.591	0.000	0.000
	5881	5888	2	1.250	9.027	-69.661	-79.233	-18.588	0.000	587.806	31.622	-9.573	-9.573	9.984	45.158	-54.730	0.000	0.000	0.000	0.000
	5882	5887	3	1.250	9.054	-99.168	-123.454	-15.896	0.000	360.572	44.087	-24.286	-24.286	11.483	30.175	-54.461	0.000	0.000	0.000	0.000
	5883	5886	4	1.250	9.081	-90.537	-129.536	-13.719	0.000	339.237	40.441	-38.999	-38.999	12.981	15.192	-54.191	0.000	0.000	0.000	0.000
	5884	5885	5	1.250	9.108	-84.639	-136.404	-12.044	0.000	317.375	37.949	-51.766	-51.766	14.284	2.156	-53.922	0.000	0.000	0.000	0.000

Appendix B-2:Active effective horizontal stress

Structural element ▲	Node ▲	Wall node ▲	Local number ▲	X ▲ [m]	Y ▲ [m]	σ'_N ▲ [kN/m ²]	σ_N ▲ [kN/m ²]	T_1 ▲ [kN/m ²]	T_2 ▲ [kN/m ²]	T_{rel} ▲ [10 ⁻³]	T_{max} ▲ [kN/m ²]	P_{water} ▲ [kN/m ²]	P_{active} ▲ [kN/m ²]	Head ▲ [m]	P_{excess} ▲ [kN/m ²]	P_{steady} ▲ [kN/m ²]	Suction ▲ [kN/m ²]	Suction _{eff} ▲ [kN/m ²]	q_n ▲ [m/day]	Q_1 ▲ [m ³ /day/m]
PositiveInterface_1_3 Element 5-66 (Interface) (Soil)	5581	5577	1	1.250	9.216	-62.423	-168.292	20.928	0.000	732.669	28.565	-105.869	-105.869	19.803	-53.026	-52.843	0.000	0.000	0.000	0.000
	5585	5578	2	1.250	9.189	-62.578	-169.948	21.218	0.000	741.128	28.630	-107.370	-107.370	19.926	-54.257	-53.113	0.000	0.000	0.000	0.000
	5586	5579	3	1.250	9.162	-62.814	-171.334	21.525	0.000	749.221	28.730	-108.519	-108.519	20.014	-55.137	-53.382	0.000	0.000	0.000	0.000
	5587	5580	4	1.250	9.135	-64.531	-174.200	21.843	0.000	741.588	29.455	-109.669	-109.669	20.102	-56.017	-53.652	0.000	0.000	0.000	0.000
	5890	5885	5	1.250	9.108	-65.182	-177.737	22.145	0.000	744.861	29.730	-112.555	-112.555	20.363	-58.634	-53.922	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-65 (Interface) (Soil)	5890	5885	1	1.250	9.108	-65.182	-177.737	22.145	0.000	744.861	29.730	-112.555	-112.555	20.363	-58.634	-53.922	0.000	0.000	0.000	0.000
	5894	5886	2	1.250	9.081	-65.065	-180.646	22.396	0.000	754.570	29.681	-115.581	-115.581	20.639	-61.389	-54.191	0.000	0.000	0.000	0.000
	5895	5887	3	1.250	9.054	-67.050	-187.189	22.529	0.000	738.187	30.519	-120.139	-120.139	21.068	-65.678	-54.461	0.000	0.000	0.000	0.000
	5896	5888	4	1.250	9.027	-78.575	-203.273	22.422	0.000	633.615	35.388	-124.698	-124.698	21.497	-69.967	-54.730	0.000	0.000	0.000	0.000
	6052	5889	5	1.250	9.000	-79.404	-204.371	17.603	0.000	492.547	35.738	-124.967	-124.967	21.497	-69.967	-55.000	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-67 (Interface) (Soil)	5557	5553	1	1.250	9.324	-61.224	-163.198	19.922	0.000	710.045	28.058	-101.974	-101.974	19.521	-50.210	-51.765	0.000	0.000	0.000	0.000
	5560	5554	2	1.250	9.297	-61.210	-164.340	20.151	0.000	718.328	28.052	-103.130	-103.130	19.610	-51.096	-52.034	0.000	0.000	0.000	0.000
	5559	5555	3	1.250	9.270	-61.673	-165.537	20.393	0.000	721.948	28.248	-103.864	-103.864	19.656	-51.560	-52.304	0.000	0.000	0.000	0.000
	5558	5556	4	1.250	9.243	-62.321	-166.918	20.651	0.000	724.047	28.521	-104.598	-104.598	19.702	-52.024	-52.574	0.000	0.000	0.000	0.000
	5581	5577	5	1.250	9.216	-62.423	-168.292	20.928	0.000	732.669	28.565	-105.869	-105.869	19.803	-53.026	-52.843	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-68 (Interface) (Soil)	5511	5087	1	1.250	9.431	-60.143	-159.359	19.064	0.000	690.712	27.601	-99.216	-99.216	19.353	-48.530	-50.686	0.000	0.000	0.000	0.000
	5517	5088	2	1.250	9.404	-60.321	-160.228	19.268	0.000	696.175	27.676	-99.908	-99.908	19.395	-48.952	-50.956	0.000	0.000	0.000	0.000
	5516	5089	3	1.250	9.377	-60.638	-161.094	19.477	0.000	700.366	27.810	-100.456	-100.456	19.423	-49.231	-51.225	0.000	0.000	0.000	0.000
	5515	5090	4	1.250	9.350	-61.147	-162.152	19.697	0.000	702.816	28.026	-101.005	-101.005	19.451	-49.510	-51.495	0.000	0.000	0.000	0.000
	5557	5553	5	1.250	9.324	-61.224	-163.198	19.922	0.000	710.045	28.058	-101.974	-101.974	19.521	-50.210	-51.765	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-69 (Interface) (Soil)	5501	5067	1	1.250	9.539	-59.137	-156.060	18.320	0.000	674.109	27.176	-96.923	-96.923	19.232	-47.315	-49.608	0.000	0.000	0.000	0.000
	5504	5068	2	1.250	9.512	-59.256	-156.854	18.495	0.000	679.315	27.226	-97.598	-97.598	19.272	-47.721	-49.877	0.000	0.000	0.000	0.000
	5503	5069	3	1.250	9.485	-59.561	-157.625	18.677	0.000	682.766	27.356	-98.064	-98.064	19.292	-47.917	-50.147	0.000	0.000	0.000	0.000
	5502	5070	4	1.250	9.458	-59.962	-158.491	18.867	0.000	685.454	27.525	-98.529	-98.529	19.311	-48.113	-50.417	0.000	0.000	0.000	0.000
	5511	5087	5	1.250	9.431	-60.143	-159.359	19.064	0.000	690.712	27.601	-99.216	-99.216	19.353	-48.530	-50.686	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-70 (Interface) (Soil)	5041	5031	1	1.250	9.647	-58.246	-153.140	17.674	0.000	659.482	26.800	-94.895	-94.895	19.137	-46.365	-48.529	0.000	0.000	0.000	0.000
	5047	5032	2	1.250	9.620	-58.423	-153.865	17.827	0.000	663.343	26.874	-95.443	-95.443	19.164	-46.644	-48.799	0.000	0.000	0.000	0.000
	5046	5033	3	1.250	9.593	-58.708	-154.571	17.985	0.000	666.241	26.995	-95.863	-95.863	19.179	-46.794	-49.069	0.000	0.000	0.000	0.000
	5045	5034	4	1.250	9.566	-59.036	-155.319	18.149	0.000	668.886	27.134	-96.283	-96.283	19.194	-46.945	-49.338	0.000	0.000	0.000	0.000
	5501	5067	5	1.250	9.539	-59.137	-156.060	18.320	0.000	674.109	27.176	-96.923	-96.923	19.232	-47.315	-49.608	0.000	0.000	0.000	0.000
PositiveInterface_1_3 Element 5-71 (Interface) (Soil)	5407	5017	1	1.250	9.755	-57.410	-150.434	17.108	0.000	646.872	26.447	-93.024	-93.024	19.057	-45.573	-47.451	0.000	0.000	0.000	0.000
	5037	5018	2	1.250	9.728	-57.530	-151.081	17.241	0.000	650.685	26.497	-93.551	-93.551	19.083	-45.831	-47.721	0.000	0.000	0.000	0.000
	5036	5019	3	1.250	9.701	-57.801	-151.755	17.381	0.000	653.135	26.612	-93.954	-93.954	19.096	-45.964	-47.990	0.000	0.000	0.000	0.000
	5035	5020	4	1.250	9.674	-58.090	-152.448	17.525	0.000	655.548	26.734	-94.358	-94.358	19.110	-46.098	-48.260	0.000	0.000	0.000	0.000
	5041	5031	5	1.250	9.647	-58.246	-153.140	17.674	0.000	659.482	26.800	-94.895	-94.895	19.137	-46.365	-48.529	0.000	0.000	0.000	0.000

Appendix B-3: Bending Moment and Shear Force of wall

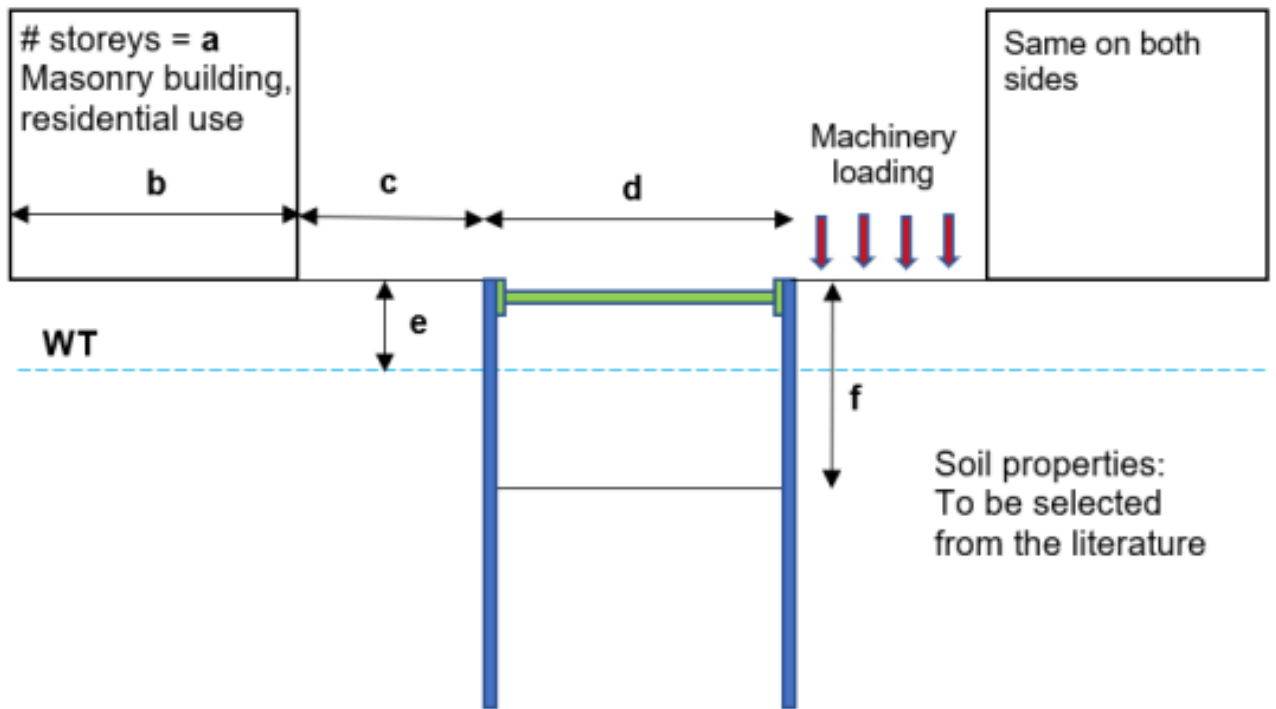
Structural element ▲	Node ▲	Local number ▲	X ▲ [m]	Y ▲ [m]	N ▲ [kN/m]	N _{min} ▲ [kN/m]	N _{max} ▲ [kN/m]	Q ▲ [kN/m]	Q _{min} ▲ [kN/m]	Q _{max} ▲ [kN/m]	M ▲ [kN m/m]	M _{min} ▲ [kN m/m]	M _{max} ▲ [kN m/m]
Plate_1_3 Element 3-77 (Plate) (wall)	5017	1	1.250	9.755	17.557	-6.750	19.145	-18.026	-18.476	0.324	10.928	-0.153	11.160
	5018	2	1.250	9.728	17.240	-6.570	18.774	-18.049	-18.493	0.317	10.442	-0.144	10.662
	5019	3	1.250	9.701	16.903	-6.388	18.384	-18.050	-18.487	0.310	9.955	-0.136	10.163
	5020	4	1.250	9.674	16.549	-6.202	17.976	-18.028	-18.458	0.303	9.469	-0.127	9.665
	5031	5	1.250	9.647	16.176	-6.014	17.549	-17.984	-18.406	0.295	8.983	-0.119	9.168
Plate_1_3 Element 3-78 (Plate) (wall)	5031	1	1.250	9.647	16.176	-6.014	17.549	-17.983	-18.405	0.295	8.983	-0.119	9.168
	5032	2	1.250	9.620	15.784	-5.822	17.103	-17.914	-18.329	0.288	8.499	-0.111	8.673
	5033	3	1.250	9.593	15.372	-5.626	16.638	-17.820	-18.228	0.280	8.018	-0.104	8.180
	5034	4	1.250	9.566	14.941	-5.426	16.153	-17.701	-18.101	0.271	7.539	-0.096	7.690
	5067	5	1.250	9.539	14.491	-5.223	15.648	-17.556	-17.948	0.263	7.063	-0.089	7.205
Plate_1_3 Element 3-79 (Plate) (wall)	5067	1	1.250	9.539	14.490	-5.223	15.648	-17.556	-17.947	0.263	7.063	-0.089	7.205
	5068	2	1.250	9.512	14.019	-5.016	15.123	-17.384	-17.767	0.254	6.592	-0.082	6.723
	5069	3	1.250	9.485	13.527	-4.805	14.577	-17.183	-17.558	0.245	6.126	-0.075	6.247
	5070	4	1.250	9.458	13.013	-4.590	14.009	-16.953	-17.319	0.236	5.666	-0.069	5.776
	5087	5	1.250	9.431	12.479	-4.370	13.419	-16.693	-17.049	0.227	5.212	-0.063	5.313
Plate_1_3 Element 3-80 (Plate) (wall)	5087	1	1.250	9.431	12.478	-4.370	13.418	-16.692	-17.047	0.227	5.212	-0.063	5.313
	5088	2	1.250	9.404	11.921	-4.145	12.806	-16.400	-16.746	0.217	4.766	-0.057	4.858
	5089	3	1.250	9.377	11.339	-3.916	12.170	-16.073	-16.408	0.208	4.328	-0.051	4.411
	5090	4	1.250	9.350	10.733	-3.681	11.508	-15.709	-16.033	0.199	3.900	-0.046	3.973
	5553	5	1.250	9.324	10.103	-3.442	10.823	-15.307	-15.619	0.190	3.482	-0.040	3.546
Plate_1_3 Element 3-81 (Plate) (wall)	5553	1	1.250	9.324	10.102	-3.441	10.821	-15.305	-15.617	0.190	3.482	-0.040	3.546
	5554	2	1.250	9.297	9.446	-3.196	10.109	-14.862	-15.160	0.181	3.075	-0.035	3.132
	5555	3	1.250	9.270	8.761	-2.944	9.367	-14.366	-14.651	0.172	2.681	-0.031	2.729
	5556	4	1.250	9.243	8.046	-2.685	8.595	-13.816	-14.085	0.163	2.301	-0.026	2.342
	5577	5	1.250	9.216	7.301	-2.420	7.793	-13.208	-13.461	0.153	1.936	-0.022	1.970
Plate_1_3 Element 3-82 (Plate) (wall)	5577	1	1.250	9.216	7.299	-2.419	7.791	-13.199	-13.452	0.153	1.936	-0.022	1.970
	5578	2	1.250	9.189	6.521	-2.147	6.955	-12.513	-12.749	0.142	1.590	-0.018	1.617
	5579	3	1.250	9.162	5.704	-1.865	6.079	-11.722	-11.938	0.128	1.262	-0.014	1.284
	5580	4	1.250	9.135	4.847	-1.574	5.163	-10.814	-11.007	0.113	0.958	-0.011	0.974
	5885	5	1.250	9.108	3.952	-1.273	4.206	-9.779	-9.947	0.097	0.680	-0.008	0.692
Plate_1_3 Element 3-83 (Plate) (wall)	5885	1	1.250	9.108	3.946	-1.272	4.200	-9.614	-9.780	0.093	0.680	-0.008	0.692
	5886	2	1.250	9.081	3.005	-0.960	3.198	-8.641	-8.785	0.090	0.434	-0.006	0.440
	5887	3	1.250	9.054	1.999	-0.634	2.129	-6.965	-7.077	0.084	0.220	-0.003	0.224
	5888	4	1.250	9.027	0.927	-0.293	0.993	-4.306	-4.374	0.065	0.067	-0.001	0.068
	5889	5	1.250	9.000	-0.211	-0.211	0.065	-0.385	-0.394	0.033	0.000	0.000	0.000

Appendix B-4: total displacement of wall

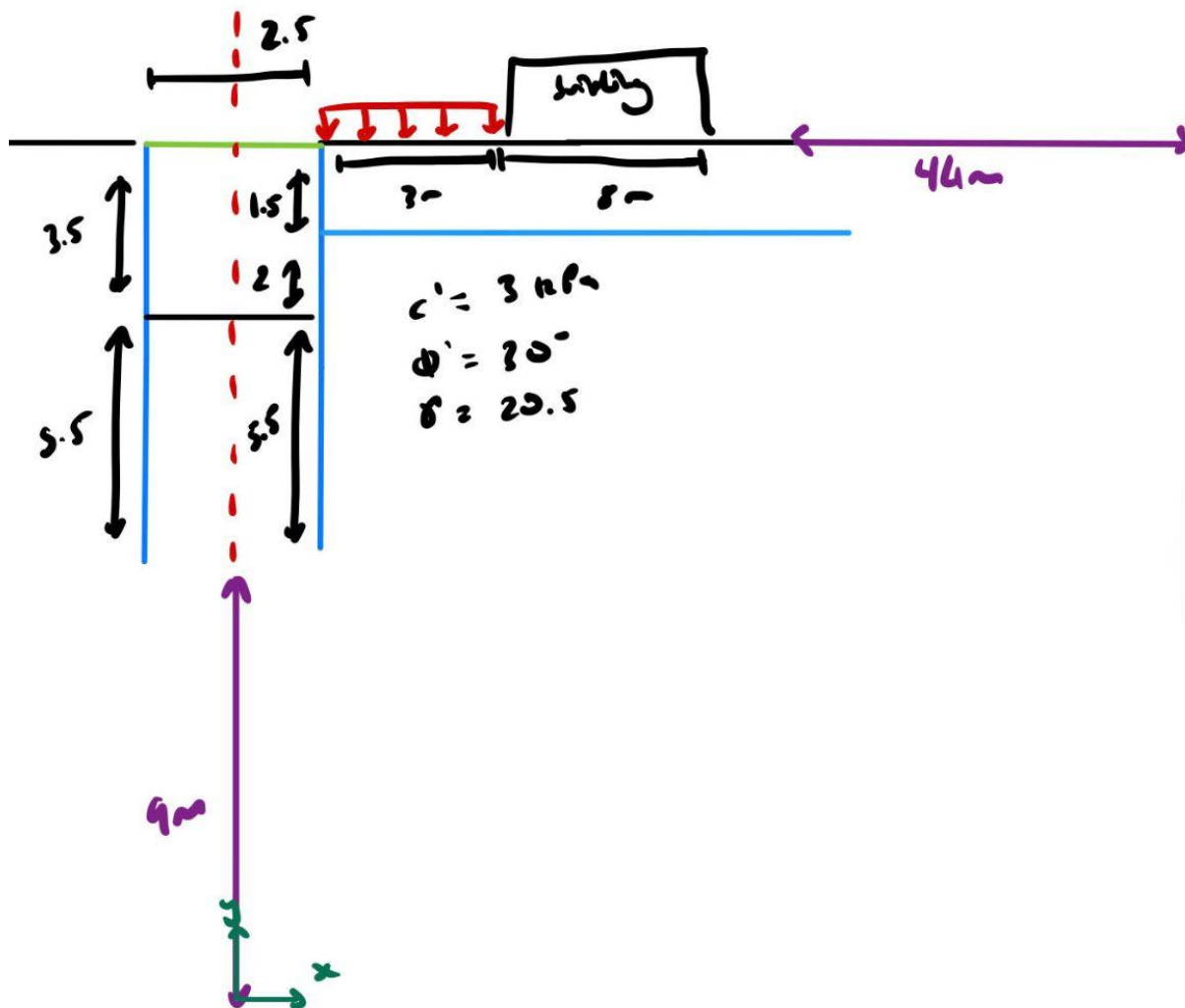
Structural element ▲	Node ▲	Local number ▲	X ▲ [m]	Y ▲ [m]	u_x ▲ [10 ⁻³ m]	u_y ▲ [10 ⁻³ m]	u ▲ [10 ⁻³ m]
Element 3-54 (Plate) (wall)	3486	2	1.250	12.208	-3.481	1.591	3.828
	3487	3	1.250	12.181	-3.453	1.591	3.802
	3488	4	1.250	12.154	-3.425	1.591	3.776
	3515	5	1.250	12.127	-3.396	1.591	3.751

Appendix C: Oversized figures

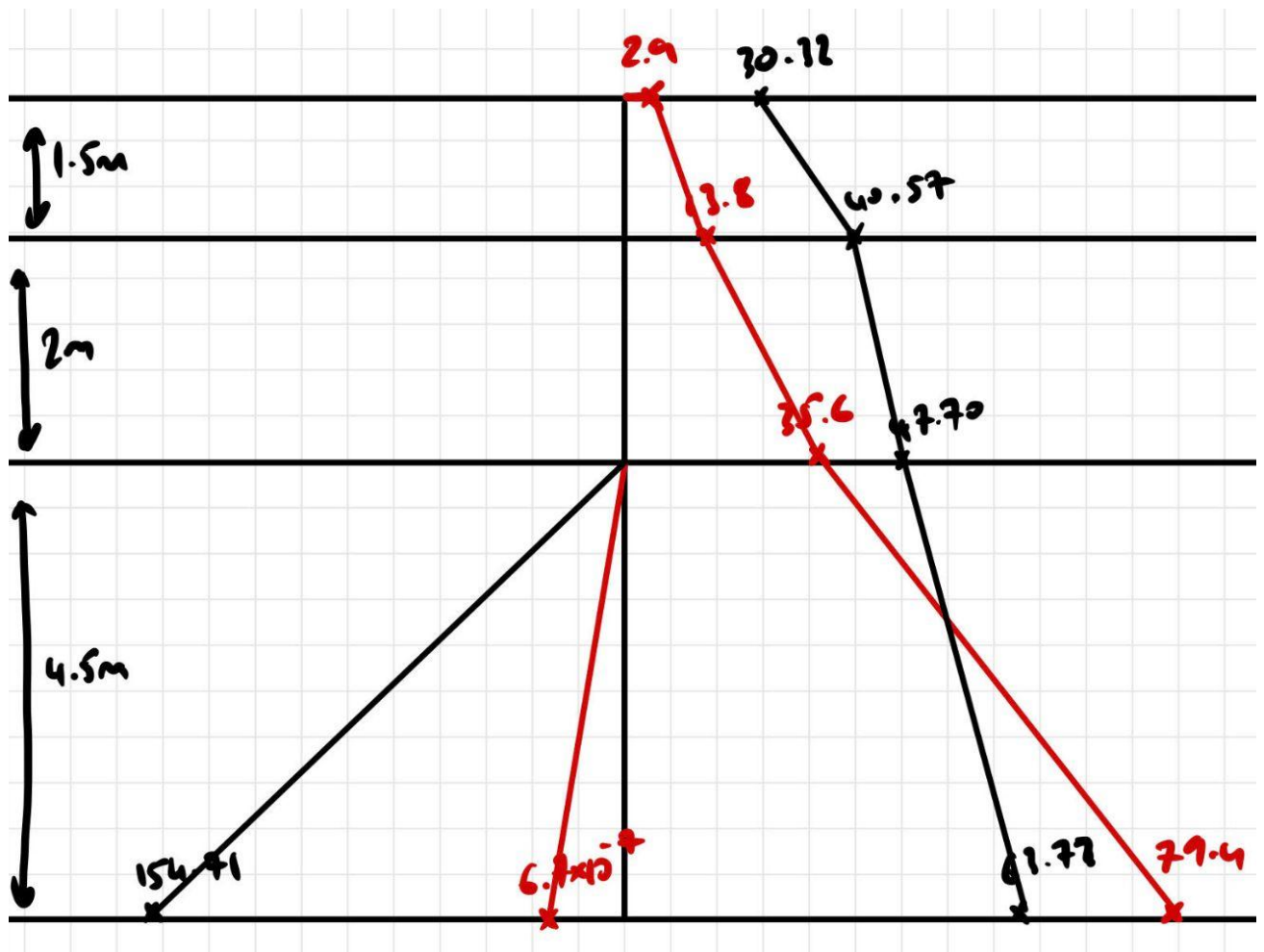
Appendix C-1: Geometry of trench



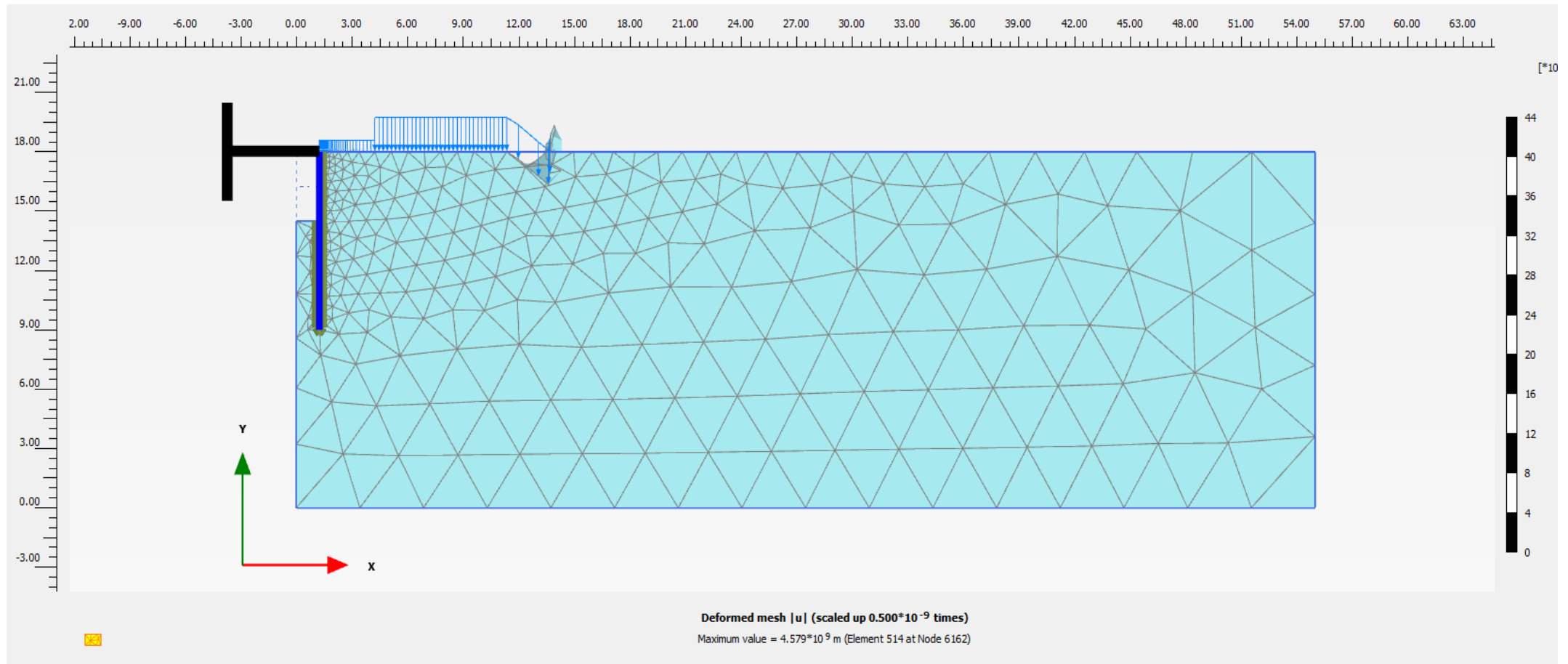
Appendix C-2: Sketched diagram of Plaxis model



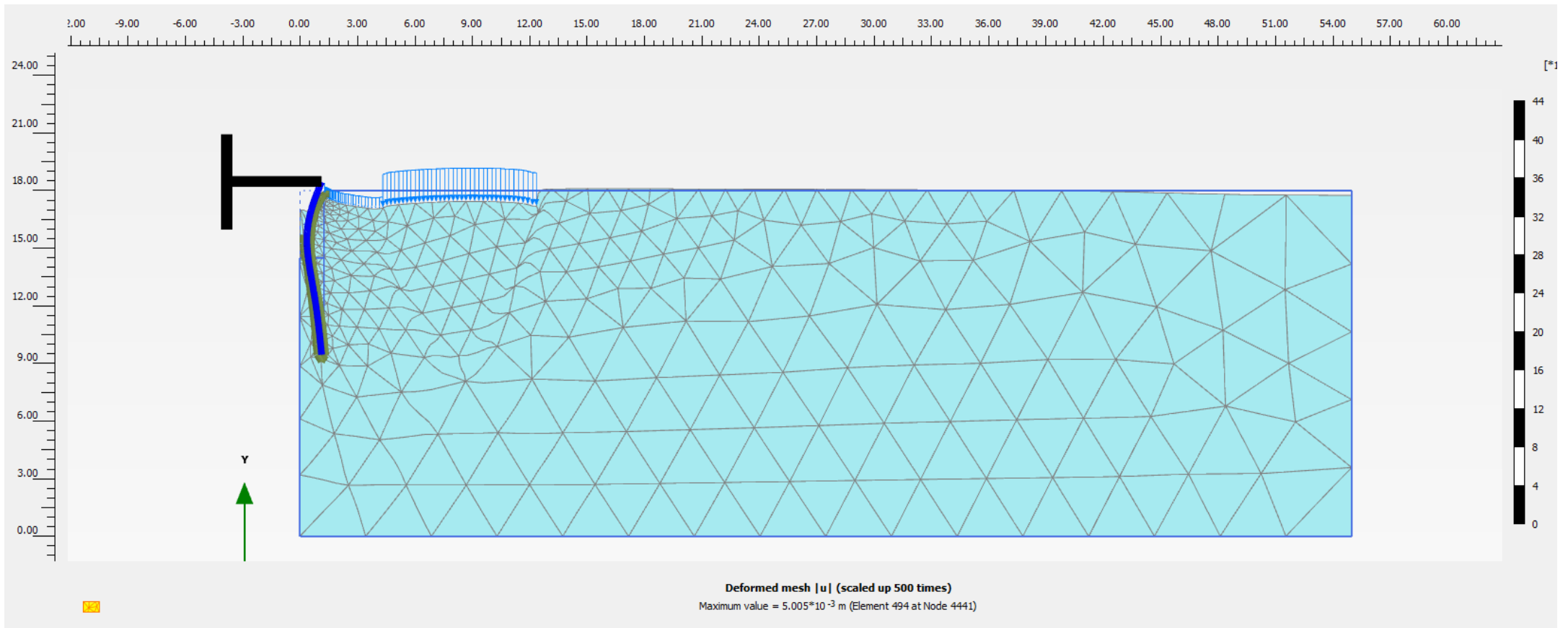
Appendix C3: Figure 6 full



Appendix C4: Figure 9 full

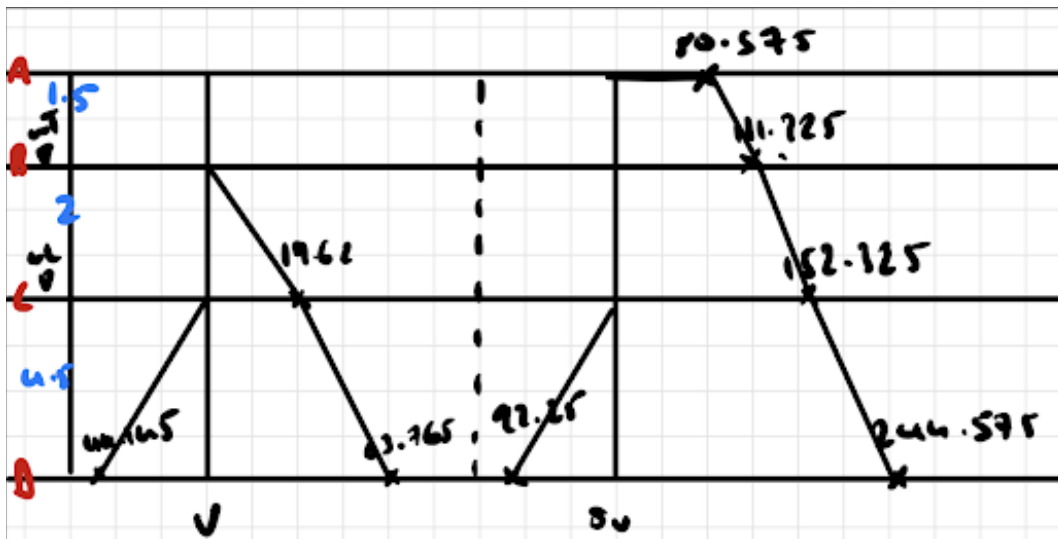


Appendix C5: Figure 10 full



Appendix D: Hand calculations and important notes

Appendix D-1: retaining wall stability analysis hand calculations



WATER PORE, u

$$u = \gamma z \text{ (under water level)}$$

where $\gamma = 9.81$

VERTICAL STRESS σ_v

$$\sigma_v = \gamma z + q, \text{ where } q = 80.575$$

$$\sigma_{vA} = 80.575$$

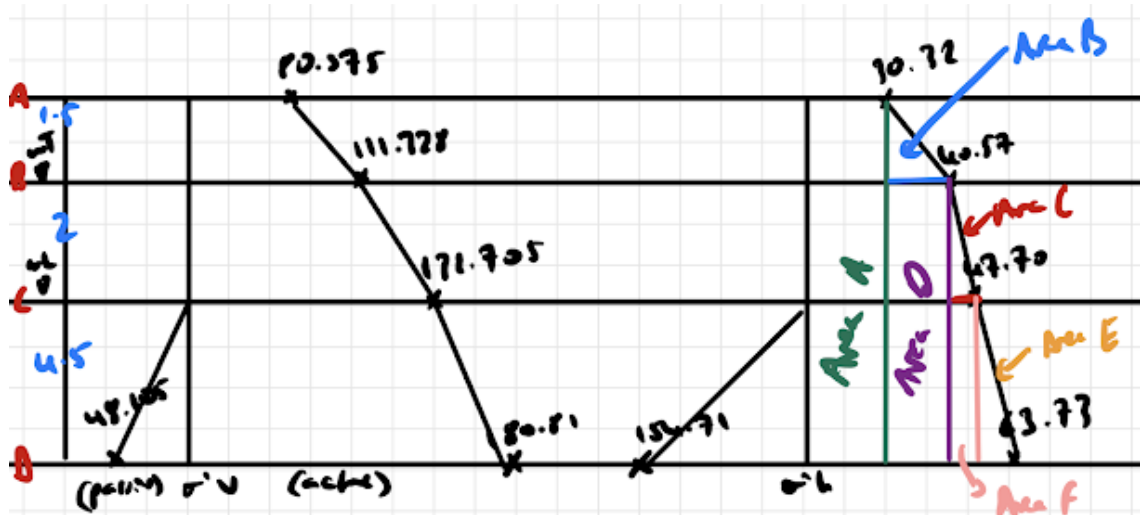
$$\sigma_{vB} = (20.50 \times 1.5) + 80.575 = 111.325$$

$$\sigma_{vC} = (20.50 \times 3.5) + 80.575 = 152.125$$

$$\sigma_{vD} = (20.50 \times 8) + 80.575$$

$$\sigma_{vD, \text{left}} = (20.50 \times 4.5) = 92.25$$

$$\hookrightarrow \sigma_v = \gamma z \text{ (no surcharge)}$$



Lever arm of Area

midsection of the shape from the point of rotation



$H/2$



$H/3$; from top $2/3$

$$I_p = H + \frac{2}{3}d - a$$

distance of top to mid of parallel side $\therefore$

$$1. 5 + 2 + \frac{1}{3}(4.5) = 6.5$$

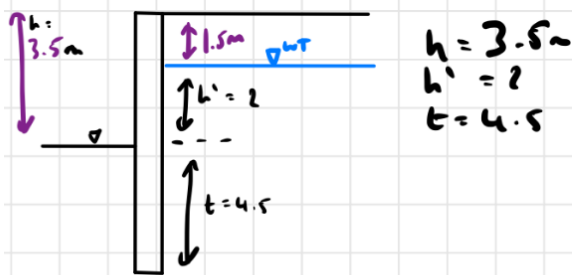
$$P_{max} = \frac{k_p \gamma d^2}{2} = \frac{3 \times 20.5 \times 4.5^2}{2} = 622.68$$

$$P_{mids} = \frac{P_{a1} \cdot d^2}{P_{max} \cdot I_p} \dots \text{all include thrust x lever arms}$$

$$P_{mids} = 31.82$$

$$f_{ad} = \frac{P_{max}}{P_{b mid}} = \frac{622.68}{31.82} \approx 20$$

Appendix D-2: HYD ULS Hydraulic failure heave two checks



$$h_r = h / (1 + \sqrt[3]{h/t} + 1) = 3.5 / (1 + \sqrt[3]{2/4.5} + 1)$$

$$h_r = 1.64$$

$$i = h_r / t = 1.64 / 4.5 = 0.365$$

$$\gamma_w \approx 90 \text{ kN/m}^3$$

See page here :

$$s_k = i \gamma_w V = 0.365 \times 10 V$$

effective weight of soil :

$$G_k = \gamma' V = 10 V$$

check 1

$$s_k \gamma_{ed,1} \leq G_k \cdot \gamma_{o,1h}$$

$$0.365 \times 10 V \times 1.35 \leq 10 V \times 0.9$$

$$4.9275 V < 9 V$$

where V is constant $\rightarrow 4.9275 < 9$ check 1 passed

check 2

$$u_k = (t + h_r) \gamma_w = (4.5 + 1.64) 9.81 = 60.2374$$

$$s_{sk,h} = t (\gamma' + \gamma_w) = 4.5 (9.81 + 9.81) = 88.29$$

$$u_k \gamma_{ed,1} \leq s_{sk,h} \gamma_{o,1h}$$

$$60.2374 \times 1.35 \leq 88.29 \times 0.9$$

$$81.3154 \geq 79.461 \quad \text{check 2 x failed}$$

2.4.7.5 Verification of resistance to failure by heave due to seepage of water in the ground

(1)P When considering a limit state of failure due to heave by seepage of water in the ground (HYD, see 10.3), it shall be verified, for every relevant soil column, that the design value of the destabilising total pore water pressure ($u_{d,t,d}$) at the bottom of the column, or the design value of the seepage force ($S_{d,t,d}$) in the column is less than or equal to the stabilising total vertical stress ($\sigma_{v,t,d}$) at the bottom of the column, or the submerged weight ($G'_{v,t,d}$) of the same column:

$$u_{d,t,d} \leq \sigma_{v,t,d} \quad (2.9a)$$

$$S_{d,t,d} \leq G'_{v,t,d} \quad (2.9b)$$

(2)P The partial factors for $u_{d,t,d}$, $\sigma_{v,t,d}$, $S_{d,t,d}$ and $G'_{v,t,d}$ for persistent and transient situations defined in A.5(1)P shall be used in equations 2.9a and 2.9b.

NOTE The values of the partial factors may be set by the National annex. Table A.17 gives the recommended values.

A.5 Partial factors for hydraulic heave limit state (HYD) verification

(1)P For the verification of hydraulic heave limit state (HYD) the following partial factors on actions (γ) shall be applied:

- $\gamma_{d,dest}$ on destabilising unfavourable permanent actions;
- $\gamma_{o,stab}$ on stabilising favourable permanent actions;
- $\gamma_{d,dest}$ on destabilising unfavourable variable actions.

NOTE The values to be ascribed to $\gamma_{d,dest}$, $\gamma_{o,stab}$ and $\gamma_{d,dest}$ for use in a country may be found in its National annex to EN 1997-1. The recommended values are given in Table A.17.

Table A.17 - Partial factors on actions (γ)

Action	Symbol	Value
Permanent		
Unfavourable ^a	$\gamma_{d,dest}$	1,35
Favourable ^b	$\gamma_{o,stab}$	0,90
Variable		
Unfavourable ^a	$\gamma_{d,dest}$	1,50
a Destabilising		
b Stabilising		

$$\gamma_w = 9.81 \text{ kN/m}^3$$

Appendix D-3: STR ULS check

ARCELORMITTAL - GÜZİN

$$\frac{b/t_f}{\epsilon} = 37 \quad (\text{class 2})$$

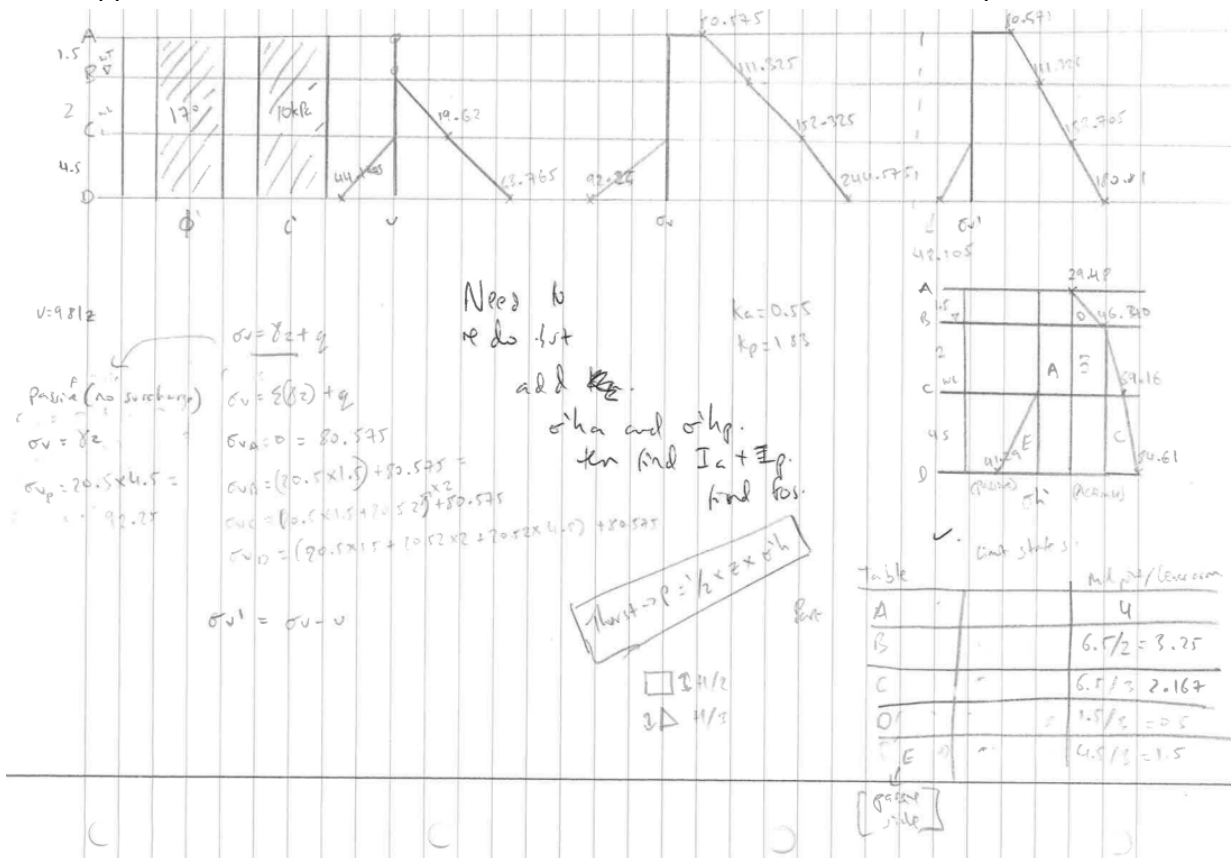
$$\frac{\left(\frac{600}{19.5}\right)}{\epsilon} = \frac{\left(\frac{600}{19.5}\right)}{0.921} = 34.76 \dots$$

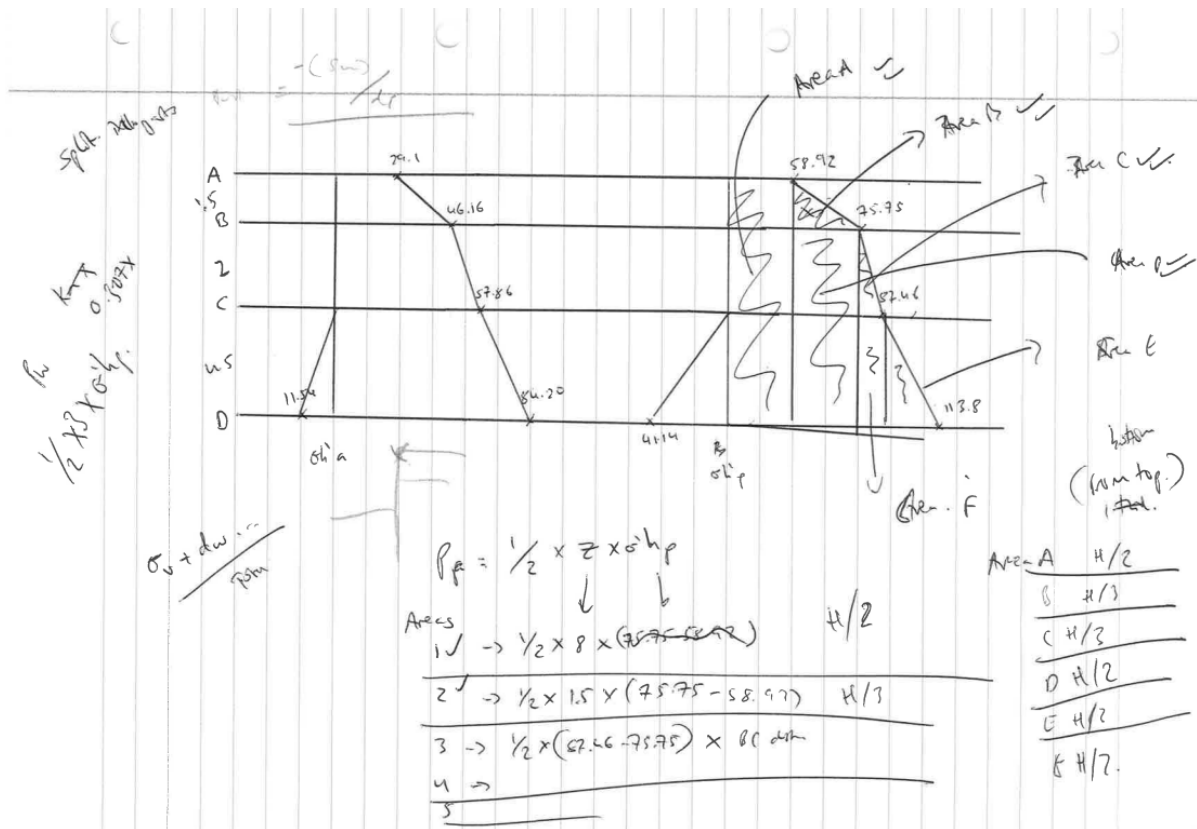
$$k = 0.933$$

$$34.76 \leq 37 \quad \checkmark$$

$$w_3 = h_y = 270$$

Appendix D-4: handwritten notes and division of areas used in excel spreadsheet





Appendix E: Eurocode 0 Tables

Appendix E-1: BS EN 1990, 2002 Table A1.1

BS EN 1990:2002+A1:2005
EN 1990:2002+A1:2005 (E)

Table A1.1 - Recommended values of ψ factors for buildings

Action	ψ_0	ψ_1	ψ_2
Imposed loads in buildings, category (see EN 1991-1-1)			
Category A : domestic, residential areas	0,7	0,5	0,3
Category B : office areas	0,7	0,5	0,3
Category C : congregation areas	0,7	0,7	0,6
Category D : shopping areas	0,7	0,7	0,6
Category E : storage areas	1,0	0,9	0,8
Category F : traffic area, vehicle weight $\leq 30\text{kN}$	0,7	0,7	0,6
Category G : traffic area, $30\text{kN} < \text{vehicle weight} \leq 160\text{kN}$	0,7	0,5	0,3
Category H : roofs	0	0	0
Snow loads on buildings (see EN 1991-1-3)*			
Finland, Iceland, Norway, Sweden	0,70	0,50	0,20
Remainder of CEN Member States, for sites located at altitude $H > 1000\text{ m a.s.l.}$	0,70	0,50	0,20
Remainder of CEN Member States, for sites located at altitude $H \leq 1000\text{ m a.s.l.}$	0,50	0,20	0
Wind loads on buildings (see EN 1991-1-4)	0,6	0,2	0
Temperature (non-fire) in buildings (see EN 1991-1-5)	0,6	0,5	0
NOTE The ψ values may be set by the National annex.			
* For countries not mentioned below, see relevant local conditions.			

6.4.3.2 Combinations of actions for persistent or transient design situations (fundamental combinations)

(1) The general format of effects of actions should be :

$$E_d = \gamma_{Sd} E \{ \gamma_{g,j} G_{k,j} ; \gamma_p P ; \gamma_{q,1} Q_{k,1} ; \gamma_{q,i} \psi_{0,i} Q_{k,i} \} \quad j \geq 1 ; i > 1 \quad (6.9a)$$

(2) The combination of effects of actions to be considered should be based on

- the design value of the leading variable action, and
- the design combination values of accompanying variable actions :

NOTE See also 6.4.3.2(4).

$$E_d = E \{ \gamma_{G,j} G_{k,j} ; \gamma_P P ; \gamma_{Q,1} Q_{k,1} ; \gamma_{Q,i} \psi_{0,i} Q_{k,i} \} \quad j \geq 1 ; i > 1 \quad (6.9b)$$

(3) The combination of actions in brackets { }, in (6.9b) may either be expressed as :

$$\sum_{j \geq 1} \gamma_{G,j} G_{k,j} "+" \gamma_P P "+" \gamma_{Q,1} Q_{k,1} "+" \sum_{i > 1} \gamma_{Q,i} \psi_{0,i} Q_{k,i} \quad (6.10)$$

or, alternatively for STR and GEO limit states, the less favourable of the two following expressions:

$$\left\{ \sum_{j \geq 1} \gamma_{G,j} G_{k,j} "+" \gamma_P P "+" \gamma_{Q,1} \psi_{0,1} Q_{k,1} "+" \sum_{i > 1} \gamma_{Q,i} \psi_{0,i} Q_{k,i} \right. \quad (6.10a)$$

$$\left. \sum_{j \geq 1} \xi_j \gamma_{G,j} G_{k,j} "+" \gamma_P P "+" \gamma_{Q,1} Q_{k,1} "+" \sum_{i > 1} \gamma_{Q,i} \psi_{0,i} Q_{k,i} \right. \quad (6.10b)$$

Where :

- "+" implies "to be combined with"
- Σ implies "the combined effect of"
- ξ is a reduction factor for unfavourable permanent actions G

Appendix E-3: BS EN 1990, 2002 Table A1.2

BS EN 1990:2002+A1:2005
EN 1990:2002+A1:2005 (E)

Table A1.2(A) - Design values of actions (EQU) (Set A)

Persistent and transient design situations	Permanent actions		Leading variable action (*)	Accompanying variable actions	
	Unfavourable	Favourable		Main (if any)	Others
(Eq. 6.10)	$\gamma_{G,j,sup} G_{k,j,sup}$	$\gamma_{G,j,inf} G_{k,j,inf}$	$\gamma_{Q,i} Q_{k,i}$		$\gamma_{Q,i} \psi_{0,i} Q_{k,i}$

(*) Variable actions are those considered in Table A1.1

NOTE 1 The γ values may be set by the National annex. The recommended set of values for γ are :

$$\gamma_{G,j,sup} = 1,10$$

$$\gamma_{G,j,inf} = 0,90$$

$$\gamma_{Q,i} = 1,50 \text{ where unfavourable (0 where favourable)}$$

$$\gamma_{Q,i} = 1,50 \text{ where unfavourable (0 where favourable)}$$

NOTE 2 In cases where the verification of static equilibrium also involves the resistance of structural members, as an alternative to two separate verifications based on Tables A1.2(A) and A1.2(B), a combined verification, based on Table A1.2(A), may be adopted, if allowed by the National annex, with the following set of recommended values. The recommended values may be altered by the National annex.

$$\gamma_{G,j,sup} = 1,35$$

$$\gamma_{G,j,inf} = 1,15$$

$$\gamma_{Q,i} = 1,50 \text{ where unfavourable (0 where favourable)}$$

$$\gamma_{Q,i} = 1,50 \text{ where unfavourable (0 where favourable)}$$

provided that applying $\gamma_{G,j,inf} = 1,00$ both to the favourable part and to the unfavourable part of permanent actions does not give a more unfavourable effect.

AC1

Appendix F: Eurocode 1 Tables

Appendix F-1: BS EN 1991-1-1, 2002 Table 6.1

Table 6.1 - Categories of use

Category	Specific Use	Example
A	Areas for domestic and residential activities	Rooms in residential buildings and houses; bedrooms and wards in hospitals; bedrooms in hotels and hostels kitchens and toilets.
B	Office areas	
C	Areas where people may congregate (with the exception of areas defined under category A, B, and D ¹⁾)	C1: Areas with tables, etc. e.g. areas in schools, cafés, restaurants, dining halls, reading rooms, receptions. C2: Areas with fixed seats, e.g. areas in churches, theatres or cinemas, conference rooms, lecture halls, assembly halls, waiting rooms, railway waiting rooms. C3: Areas without obstacles for moving people, e.g. areas in museums, exhibition rooms, etc. and access areas in public and administration buildings, hotels, hospitals, railway station forecourts. C4: Areas with possible physical activities, e.g. dance halls, gymnastic rooms, stages. C5: Areas susceptible to large crowds, e.g. in buildings for public events like concert halls, sports halls including stands, terraces and access areas and railway platforms.
D	Shopping areas	D1: Areas in general retail shops D2: Areas in department stores
¹⁾ Attention is drawn to 6.3.1.1(2), in particular for C4 and C5. See EN 1990 when dynamic effects need to be considered. For Category E, see Table 6.3 NOTE 1 Depending on their anticipated uses, areas likely to be categorised as C2, C3, C4 may be categorised as C5 by decision of the client and/or National annex. NOTE 2 The National annex may provide sub categories to A, B, C1 to C5, D1 and D2 NOTE 3 See 6.3.2 for storage or industrial activity		

Appendix F-2: BS EN 1991-1-1, 2002 Table 6.2

Table 6.2 - Imposed loads on floors, balconies and stairs in buildings

Categories of loaded areas	q_k [kN/m ²]	Q_k [kN]
Category A		
- Floors	1,5 to <u>2,0</u>	<u>2,0</u> to 3,0
- Stairs	<u>2,0</u> to 4,0	<u>2,0</u> to 4,0
- Balconies	<u>2,5</u> to 4,0	<u>2,0</u> to 3,0
Category B	2,0 to <u>3,0</u>	1,5 to <u>4,5</u>
Category C		
- C1	2,0 to <u>3,0</u>	3,0 to <u>4,0</u>
- C2	3,0 to <u>4,0</u>	2,5 to 7,0 (<u>4,0</u>)
- C3	3,0 to <u>5,0</u>	<u>4,0</u> to 7,0
- C4	4,5 to <u>5,0</u>	3,5 to <u>7,0</u>
- C5	<u>5,0</u> to 7,5	3,5 to <u>4,5</u>
category D		
- D1	<u>4,0</u> to 5,0	3,5 to 7,0 (<u>4,0</u>)
- D2	4,0 to <u>5,0</u>	3,5 to <u>7,0</u>

Appendix F-3: BS EN 1991-1-1, 2002 Table 6.5

(1) Forklifts should be classified in 6 classes FL 1 to FL 6 depending on net weight, dimensions and hoisting loads, see Table 6.5.

Table 6.5 - Dimensions of forklift according to classes FL

Class of Forklift	Net weight [kN]	Hoisting load [kN]	Width of axle a [m]	Overall width b [m]	Overall length l [m]
FL 1	21	10	0,85	1,00	2,60
FL 2	31	15	0,95	1,10	3,00
FL 3	44	25	1,00	1,20	3,30
FL 4	60	40	1,20	1,40	4,00
FL 5	90	60	1,50	1,90	4,60
FL 6	110	80	1,80	2,30	5,10

(2) The static vertical axle load Q_k of a forklift depends on the forklift classes FL1 to FL6 and should be obtained from Table 6.6.

Appendix G: Eurocode 7 Tables

Appendix G-1: BS EN 1997-1, 2004 Table A.3

Table A.3 - Partial factors on actions (γ_F) or the effects of actions (γ_E)

Action		Symbol	Set	
			A1	A2
Permanent	Unfavourable	γ_G	1,35	1,0
	Favourable		1,0	1,0
Variable	Unfavourable	γ_Q	1,5	1,3
	Favourable		0	0

Appendix G-2: BS EN 1997-1, 2004 Table A.4

Table A.4 - Partial factors for soil parameters(γ_M)

Soil parameter	Symbol	Set	
		M1	M2
Angle of shearing resistance ^a	$\gamma_{\phi'}$	1,0	1,25
Effective cohesion	$\gamma_{c'}$	1,0	1,25
Undrained shear strength	γ_{c_u}	1,0	1,4
Unconfined strength	γ_{q_u}	1,0	1,4
Weight density	γ_f	1,0	1,0
^a This factor is applied to $\tan \phi'$			

Appendix G-3: BS EN 1997-1, 2004 Table A.5

Table A.5 - Partial resistance factors (γ_R) for spread foundations

Resistance	Symbol	Set		
		$R1$	$R2$	$R3$
Bearing	$\gamma_{R,v}$	1,0	1,4	1,0
Sliding	$\gamma_{R,h}$	1,0	1,1	1,0

Appendix H: ArcelorMittal GU section property table

GU[®] sections

Section	Dimensions				A	G _{sp}	G _w	I _y	W _{el,y}	S _y	W _{pl,y}
	b	h	t _f	t _w							
	mm	mm	mm	mm							
GU 6N	600	309	6.0	6.0	89	41.9	70	9 670	625	375	765
GU 7N	600	310	6.5	6.4	94	44.1	74	10 450	675	400	825
GU 7S	600	311	7.2	6.9	98	46.3	77	11 540	740	440	900
GU 7HWS	600	312	7.3	6.9	101	47.4	79	11 620	745	445	910
GU 8N	600	312	7.5	7.1	103	48.5	81	12 010	770	460	935
GU 8S	600	313	8.0	7.5	108	50.8	85	12 800	820	490	995
GU 10N	600	316	9.0	6.8	118	55.8	93	15 700	990	565	1 160
GU 11N	600	318	10.0	7.4	128	60.2	100	17 450	1 095	630	1 280
GU 12N	600	320	11.0	8.0	137	64.6	108	19 220	1 200	690	1 400
GU 13N	600	418	9.0	7.4	127	59.9	100	26 590	1 270	755	1 535
GU 14N	600	420	10.0	8.0	136	64.3	107	29 410	1 400	830	1 685
GU 15N	600	422	11.0	8.6	146	68.7	115	32 260	1 530	910	1 840
GU 16N	600	430	10.2	8.4	154	72.6	121	35 950	1 670	980	1 988
GU 18N	600	430	11.2	9.0	163	76.9	128	38 650	1 800	1 055	2 134
GU 20N	600	430	12.2	9.5	172	81.1	135	41 320	1 920	1 125	2 280
GU 21N	600	450	11.1	9.0	174	81.9	137	46 380	2 060	1 195	2 422
GU 22N	600	450	12.1	9.5	183	86.1	144	49 460	2 200	1 275	2 580
GU 23N	600	450	13.1	10.0	192	90.4	151	52 510	2 335	1 355	2 735
GU 27N	600	452	14.2	9.7	207	97.4	162	60 580	2 680	1 525	3 087
GU 28N	600	454	15.2	10.1	216	101.8	170	64 460	2 840	1 620	3 269
GU 30N	600	456	16.2	10.5	226	106.2	177	68 380	3 000	1 710	3 450
GU 31N	600	452	18.5	10.6	233	109.9	183	69 210	3 065	1 745	3 525
GU 32N	600	452	19.5	11.0	242	114.1	190	72 320	3 200	1 825	3 687
GU 33N	600	452	20.5	11.4	251	118.4	197	75 410	3 340	1 905	3 845
GU 16-400	400	290	12.7	9.4	197	62.0	155	22 580	1 560	885	1 815
GU 18-400	400	292	15.0	9.7	221	69.3	173	26 090	1 785	1 015	2 080